

**Committing to Implementation Intentions: Attention and  
Memory Effects for Selected Situational Cues**

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Running Head: ATTENTION AND MEMORY EFFECTS FOR SELECTED SITUATIONAL CUES

**Committing to Implementation Intentions:**  
**Attention and Memory Effects for Selected Situational Cues**

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## Abstract

Two studies investigated the assumption that forming implementation intentions (Gollwitzer, 1993, 1999) results in a lasting but commitment-dependent heightened activation of situational cues specified in these plans. In Study 1, cues specified in the if-components of implementation intentions attracted more attention than non-specified cues when presented in a dichotic listening task. This effect did not hold for control participants for whom the specified cues were only made salient by a familiarization procedure. In Study 2, the recall of cues specified in the if-components of implementation intentions was observed to be better than that of non-specified cues both 15 min after forming the implementation intentions and after a delay of two days. In addition, it was discovered that both commitment to the respective goal and the formed implementation intentions moderated this effect. Cues specified in implementation intentions were better remembered than non-specified cues independent of the delay, but only if commitment was high.

**KEYWORDS:** intentions, goals, implementation intentions, cognitive accessibility, intention superiority effect

**Committing to Implementation Intentions:**

**Attention and Memory Effects for Selected Situational Cues**

There is strong evidence that intentions (i.e., behavioral goals) have a special status in long-term memory, showing greater cognitive accessibility than other memory contents (e.g., Goschke & Kuhl, 1993; Koriat, Ben-Zur, & Nussbaum, 1990; Marsh, Hicks, & Bink, 1998; Marsh, Cook, Meeks, Clark-Foos, & Hicks, 2007). This special status is assumed to be a result of a heightened activation level (i.e., above baseline activation) of intentions compared to other information. Goschke and Kuhl (1993) have proposed a persistence hypothesis whereby increased activation is described in terms of the persistence of the memory trace (i.e., resistance to decay or interferences). The effect that memories for intended actions are stored in a heightened state of activation or otherwise afforded a privileged status in memory has been referred to as ISE (intention-superiority effect; Goschke & Kuhl, 1993). This heightened state of activation prior to the completion of the intention was observed by means of recognition tasks (e.g., Goschke & Kuhl, 1993), lexical decision tasks (e.g., Marsh et al., 1998; Marsh, Hicks, & Bryan, 1999), recall tasks (Koriat et al., 1990; Penningroth, 2005), and event-based prospective memory paradigms (Kazén, Kaschel, & Kuhl, 2008, Study 1). Badets, Blandin, and Bouquet (2006) were even able to show the special status of intentions in long-term memory in the domain of motor skill learning. The ISE has been observed with assigned and everyday intentions that were long-term (up to 1 year) and real for the participants (e.g., Kuhl & Goschke, 1994; Maylor, Darby, & Della Sala, 2000). Some authors (e.g., Dockree & Ellis, 2001; Marsh et al., 1999) even observed that after an intention has been fulfilled or cancelled, it seems to be inhibited, at least relative to neutral material. This inhibition effect may be due to active inhibition or decay (Marsh et al., 1998, 1999).

### Research on Implementation Intentions

Gollwitzer (1993, 1999) postulated a distinction between two types of intentions, goal intentions versus implementation intentions. Whereas goal intentions specify an intended behavior or outcome, implementation intentions specify a cue (an external situation or an inner state; if-component) that is then linked to a behavior that is instrumental to reaching the goal described in the goal intention (i.e., a goal-directed response; then-component; Gollwitzer, 1993, 1999; Achtziger, Gollwitzer, & Sheeran, 2008). Thereby, a strong link between the if- and the then-component is established that generates the positive effects of these action plans on goal attainment. Research on implementation intentions has shown (summaries by Achtziger & Gollwitzer, 2008; Gollwitzer & Sheeran, 2006) that implementation intentions formed in the service of goal intentions enhance goal attainment. Action initiation in the presence of the specified cue is observed to be immediate (Gollwitzer & Brandstätter, 1997), efficient (Brandstätter, Lengfelder & Gollwitzer, 2001), and does not require further conscious intent (Bayer, Achtziger, Gollwitzer, & Moskowitz, 2009).

Inspired by the findings of ISE research, implementation intention research explored whether specifying critical cues in the if-component of implementations intentions causes the representation of these cues to become highly activated and thus more accessible than alternative cues. There are some findings in line with this assumption. For instance, Webb & Sheeran (2004, Study 1) showed that implementation intentions increase the likelihood to detect a specified cue. The implementation intention was geared at supporting the detection of the letter *F* in a text, and accessibility was indicated by the number of capital *F*s participants could detect. Implementation intention participants were better in detecting the letter *F* than control participants. It was concluded that forming an implementation intention heightens the cognitive activation of the if-component, and consequently the specified cue is more easily detected compared to other cues. Also, Parks-Stamm, Gollwitzer, and Oettingen (2007, Study 1) discovered that specifying cues by forming implementation intentions leads to an increase of their accessibility accompanied by a

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decrease of the accessibility of alternative cues. In an identification task, implementation intention participants were better at recognizing the cues specified in their implementation intentions than control participants. Their identification performance for alternative, non-specified cues was lower than that of control participants.

Whereby these studies assessed heightened activation in terms of perceptual readiness, the present studies examine whether cues specified in the if-component of implementation intentions are also more readily attended to (Study 1) and better remembered than alternative cues (Study 2). In addition, we analyze both commitment to the superordinate goal intention and commitment to enact the formed implementation intention as moderators of these effects (Study 2).

**The Present Research**

Study 1 investigated whether cues specified in implementation intentions draw more attention than alternative, non-specified cues. As experimental paradigm a dichotic listening task was used (e.g., Cherry & Taylor 1954; Nielson & Sarason, 1981; Treisman, 1960). We predicted that cues specified in implementation intentions will draw more attention in a dichotic listening task than alternative cues. In yoked control participants who familiarized themselves with the critical cues, no such attention drawing effects should be observed. Moreover, implementation intention participants should show stronger attention drawing effects on specified cues than control participants. These predictions were based on the assumption that a heightened activation of the critical cues should only be generated by specifying them in the if-components of implementation intentions, whereas familiarizing oneself with these cues should not suffice. Study 2 examined the assumption that specifying cues in the if-component of implementation intentions results in a better recall of these cues compared to alternative, non-specified cues. However, if the superordinated goal intention is deactivated or if participants are only weakly committed to perform their implementation intentions, differences between critical and alternative cues should no longer be observed in a recall test as the heightened accessibility should have

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3 vanished. To test these hypotheses, the recall of cues specified in implementation intentions was  
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5 tested 15 min after the formation of the implementation intentions and after a delay of one day.  
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7 The delayed recall test was conducted to critically test the assumption that the state of heightened  
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9 activation persists over time.  
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### 11 12 **Study 1: Disruption of Focused Attention** 13

14 Participants were invited to take part in allegedly two studies. In the first study,  
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16 implementation intention participants were asked to name a goal which they intended to realize  
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18 within the next three weeks. They formed this goal and planned its realization by forming several  
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20 implementation intentions filling out a questionnaire. Each control participant was matched to an  
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22 implementation intention participant (i.e., yoked). Control participants were required to judge the  
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24 layout of the implementation intentions questionnaire and to write down the words describing  
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26 their yoked partners' implementation intentions, this way increasing familiarity with these words.  
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31 The allegedly second study was run the next day. All participants were asked to perform  
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33 a dichotic listening task. They repeated (i.e., "shadowed") words presented to the attended  
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35 channel by means of headphones. Words presented to this channel were not associated with the  
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37 words specified in the if-component of implementation intention participants' plans. Some of the  
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39 words presented to the unattended channel, however, were taken from the if-components of the  
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41 implementation intention. These words qualified as the critical targets. Other words also presented  
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43 to the unattended channel were not related to the if-components of the implementation intentions  
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45 (i.e., non-critical targets). Words presented to the unattended channel were also presented by  
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47 means of the headphones, using the same ITI as words presented to the attended channel. As  
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49 dependent variables, pronunciation errors and latencies for the targets presented to the attended  
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51 channel were recorded.  
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56 In the implementation intention condition, it was predicted that the heightened activation  
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58 of the critical cues caused by forming implementation intentions should interfere with shadowing  
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60 the words presented to the attended channel. Consequently, a shift of attention from the targets

presented to the attended channel to the critical targets presented to the unattended channel was expected in this condition. This shift should be indicated by slower shadowing latencies (e.g., Lewis, 1970; Treisman, 1974) and more errors (e.g., Dawson & Schell, 1982; Nielson & Sarason, 1981) whenever critical cues but not noncritical targets were presented to the unattended channel. No such effects were expected in the control condition.

**Method**

**Participants**

Male University students participated in the experiment (n = 32; age:  $M = 24.91$ ;  $SD = 3.17$ ). Participants were enrolled in different fields of studies. They were recruited by telephone for (allegedly) two different experiments and randomly assigned to the two between conditions.

**Design**

The study followed a 2 between (Intention: control vs. implementation intention)  $\times$  2 within (Target Words: critical vs. noncritical) mixed-model design. For counterbalancing reasons, participants were randomly assigned to a left vs. right ear condition (i.e., the target words were either presented to the left or right ear). As dependent variables pronunciation latencies and pronunciation errors of words presented to the attended channel were measured.

**Procedure**

Participants were told that the two studies would be scheduled 24 hours apart. The first study was described as a study about how people organize their everyday life, the second study as investigating auditory vigilance. Different experimenters were named as responsible for the two studies to confirm that the two studies would be independent from each other.

For the (allegedly) first study, participants arrived in the laboratory in groups of 3 to 6. In the implementation intention condition, participants named a goal that they intended to realize within the next months on a questionnaire. They were told that this goal should fulfill the following criteria: 1) it should concern an issue that they already had considered for a while, 2) it should be possible to break down its realization into separate steps, and 3) the outcomes of the



goal attainment should be highly appreciated. After implementation intention participants named their goal, they answered five questions about their commitment to this goal on nine point scales (0 = not important/not at all to 9 = very important/very much).

Afterwards, implementation intention participants were told that they should plan the implementation of their goal and were given an example on how to do this (e.g., planning a trip to Australia). Participants studied this example and then planned the implementation of their own goal. They named five goal-directed behaviors suited to support the realization of their goal and listed them on a sheet of paper. For each of these behaviors, participants were asked to form implementation intentions by writing down where and when (i.e., the if-components) they planned to perform each of these behaviors. After finishing this task, participants answered questions about their commitment to the goal and to their implementation intentions (see commitment variables below).

Each participant in the control condition was randomly assigned to one of the implementation intention participants (i.e., yoked design). They received the completed questionnaire of their yoked implementation intention partner. Control participants were asked to judge the questionnaire's layout and to copy the words the yoked implementation intention participants had used to form their implementation intentions. Thereby, control participants were made familiar with the words used by implementation intention participants. Thus, we controlled for salience effects of the critical words in the dichotic listening task. This procedure was quite similar to that used in studies examining the ISE (e.g., Marsh et al., 2007, Study 1 and 2).

Next day, participants were greeted by another experimenter conducting the dichotic listening task. They were seated in individual cubicles where the words were presented by means of headphones that were connected to a computer. Participants were told that words would be presented simultaneously to their left and right ears, and that they would have to repeat as quickly as possible the words that would be presented on either their left (or right) ear (attention to the channels was counterbalanced). All participants were told to be as fast and accurate as possible in

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repeating the words on the shadowed channel. The computer recorded pronunciation latencies on the presented words of the shadowed channel and pronunciation errors. Latencies were recorded from the beginning of the presentation of the word until the beginning of the response. No responses and wrong responses (i.e., pronouncing a different word, pronouncing the word only partially or jolty) were defined as errors.

Simultaneously to the dichotic listening task, participants performed a probe reaction time task, in which they pressed a button as quickly as possible whenever a light went on at random times. We emphasized that this task was only secondary in importance (it was installed to prevent participants from listening to the unattended channel). Then, the headphones were mounted and the microphones were adjusted. The experimenter went to the control room and gave instructions by means of an intercom system. All participants were told there would be five word lists to repeat, starting with a practice list. After the dichotic listening task had been finished, participants were carefully debriefed, thanked, and paid (10 Euros).

**Material**

**Target words.** For each implementation intention participant, 10 words (nouns) were randomly selected from their implementation intention questionnaire. They served as critical words describing the if-components of their implementation intentions in the dichotic listening task. For this task, two matched word lists were constructed, consisting of 98 nouns each. Words were taken from Ruoff’s (1981) “*Häufigkeitwörterbuch gesprochener Sprache*” (German Dictionary of Word Frequencies) and had one to six syllables. Only words that were evaluated as neither positive nor negative were selected (e.g., countryside, inhabitants). Word pairs were constructed by matching the critical words with words that had the same number of syllables and frequency of occurrence. Control participants were presented the same words as their yoked implementation intention partners.

Word lists were prepared in advance, with the exception of the critical words, which were inserted separately for each participant prior to the experiment. Each word was presented in the

same volume and length for each participant, and both words of a word pair were presented at the same moment.

**Trials.** Word pairs were presented on five lists, with a 3 sec break between them. At the onset of the presentation of each list “Achtung” (“attention”) was simultaneously presented to both ears. The ITI between the presentation of the words was 1 sec. We ensured that the lists were highly similar regarding the mean length and frequency of the stimulus words. The first list was a practice list of 30 word pairs. The other four lists consisted of 17 word pairs each, whereby in the first and third list critical targets were interspersed (i.e., critical lists). Both critical lists contained 10 critical targets and 7 non-critical targets that were presented to the unattended channel. The critical targets were presented in a randomized order.

**Commitment variables.** After implementation intention participants had finished the first part of the experiment (the allegedly Study 1), they answered the following questions: 1) How important is this goal compared to your other goals at the moment?, 2) How strongly are you determined to realize your goal?, 3) How strongly are you committed to realize your goal?, and 4) How important is the attainment of your goal to you? Concerning their implementation intentions, implementation intention participants answered the question “How strongly are you committed to perform the listed goal-directed behaviors as soon as the specified situations occur?” All answers were given on nine point scales ranging from 0 (= not at all) to 9 (= strongly).

Control participants did not answer these questions as they were not required to plan out the implementation of a goal (see above).

## Results

### Probe Reaction Times

There were no differences between probe reaction times in the implementation intention and the control condition,  $F < 1$ . Because this task should guarantee that participants were focused on other tasks than listening to the unattended channel, there were no predictions concerning this variable.

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**Commitment Variables**

A correlation analysis revealed that answers to the goal commitment questions (see above) were highly inter-correlated (Cronbach’s  $\alpha = .79$ ). The overall mean on the answers to these questions was high ( $M = 6.58$ ;  $SD = 1.41$ ). Commitment to the implementation intentions was high as well ( $M = 6.35$ ,  $SD = 1.28$ ).

**Outliers**

Pronunciation latencies were screened for outliers; latencies of less than 400 ms and more than 2000 ms were excluded. Next, latencies of wrong responses (i.e., pronouncing a different word or pronouncing a word partially or jolty) and latencies of more than 3 standard deviations above/below mean were excluded from the data set. Participants who had more than 20% missing data due to these exclusions were eliminated from the data analyses (i.e., 3 participants, leaving a total of 29 participants in the data set). Testing whether there were effects on the location of the attended channel (left vs. right) revealed no significant effects (all  $F$ s  $< 1$ ).

**Shadowing Latencies**

For each participant, mean pronunciation latencies on the shadowed words were computed separately for trials in which critical and noncritical targets were presented to the unattended channel. A 2 within (Target Words: critical vs. noncritical)  $\times$  2 between (Intention: control vs. implementation intention) mixed-model ANOVA on these latencies revealed a significant Target Words  $\times$  Intention interaction effect,  $F(1,27) = 5.06$ ,  $p < .05$ ,  $\eta^2 = .16$ . The significant main effect of Target Words indicated that latencies on the shadowed words were generally longer when critical targets compared to noncritical targets were presented to the unattended channel,  $F(1,27) = 7.22$ ,  $p < .05$ ,  $\eta^2 = .21$ . The factor Intention did not reach significance,  $F(1,27) = 2.39$ ,  $p = .13$ ,  $\eta^2 = .08$  (see Figure 1).

As predicted, planned  $t$ -tests revealed that shadowing latencies were slower when critical targets were presented to the unattended channel in the implementation intention condition ( $M = 1120.63$ ,  $SD = 124.19$ ) compared to the control condition ( $M = 1039.93$ ,  $SD = 93.89$ ),  $t(27) =$

1.93,  $p = .05$ . However, there was no difference between implementation intention ( $M = 1058.09$ ,  $SD = 86.53$ ) and control participants ( $M = 1034.36$ ,  $SD = 78.96$ ) on shadowing latencies when noncritical targets were presented to the unattended channel,  $t < 1$ . Moreover, implementation intention participants' shadowing was slower when critical targets compared to noncritical targets were presented to the unattended channel,  $t(13) = 3.05$ ,  $p < .01$ . This effect was not observed for control participants,  $t < 1$ .

### Errors

A 2 within (Target Words: critical vs. noncritical)  $\times$  2 between (Intention: control vs. implementation intention) mixed-model ANOVA on the number of errors in the shadowing task revealed a significant Target Words  $\times$  Intention interaction effect,  $F(1,27) = 4.22$ ,  $p = .05$ ,  $\eta^2 = .14$ . No significant main effects of Target Word and Intention were observed (all  $F_s < 1$ ).

Planned  $t$ -tests revealed that implementation intention participants made more errors in the shadowing task if critical targets ( $M = 1.71$ ,  $SD = 1.49$ ) were presented to the unattended channel compared to noncritical targets ( $M = 0.96$ ,  $SD = 0.84$ ),  $t(13) = 2.24$ ,  $p < .05$ ; this difference was not significant for control participants ( $M = 1.37$ ,  $SD = 1.20$  vs.  $M = 1.13$ ,  $SD = 0.74$ ),  $t < 1$ . Implementation intention participants tended to make more errors in the shadowing task ( $M = 1.71$ ,  $SD = 1.49$ ) compared to control participants ( $M = 1.13$ ,  $SD = 0.74$ ) if critical words were presented to the unattended channel,  $t(27) = 1.34$ ,  $p < .10$  (*one-tailed*); the difference between implementation intention ( $M = 0.96$ ,  $SD = 0.84$ ) and control participants ( $M = 1.37$ ,  $SD = 1.20$ ) for non-critical words was reversed, albeit it did not reach significance,  $t(27) = 1.03$ ,  $p < .15$  (*one-tailed*).

### Discussion

We investigated whether cues specified in the if-component of implementation intentions attract attention despite strong distractions. These specified cues were presented to the unattended channel in a dichotic listening task while participants repeated words presented to the attended channel. Shadowing latencies were assessed as an indicator of attention being attracted to the

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unattended channel. In the implementation intention condition, the presentation of specified cues to the unattended channel led to significantly longer shadowing latencies compared to the presentation of non-critical targets. No such effect was observed with control participants who had been made familiar with the critical cues due to a yoked familiarization manipulation. Implementation intention participants also showed longer shadowing latencies than control participants if critical cues were presented to the unattended channel. However, there was no difference between implementation intention and control participants concerning pronunciation latencies if non-critical targets were presented to the unattended channel. The observed pattern of latency results strongly suggests that forming implementation intentions make cues specified in the if-components attract attention. This effect can not be explained by implementation intention participants becoming consciously aware of their plans (what might have been the case due to the increased presentation of the critical targets compared to noncritical targets), because an increased awareness of the critical targets should have caused some general slowing on noncritical targets as well as on the probe reaction task. However, neither of these results has been obtained.

The assessed pronunciation errors were also in line with our predictions. Implementation intention participants significantly revealed a greater number of shadowing errors if critical targets were presented to the unattended channel compared to noncritical targets. This result was not observed in the control condition. In addition, implementation intention participants tended to make more errors in the shadowing task if critical targets were presented to the unattended channel than control participants, whereas control participants tended to make more shadowing errors when noncritical targets were presented to the unattended channel.

It is important to note that these findings were obtained by using words as critical targets that were idiosyncratically specified by the implementation intention participants themselves. This attests strong ecological validity of the present findings. Moreover, our findings suggest that the attention drawing effect of cues specified in implementation intentions is quite strong. Early experiments on dichotic listening (e.g., Cherry, 1953; Moray, 1959) observed that only very little

is processed of words presented to the unattended channel. Participants often cannot report any of the words presented to the unattended channel, even if these words have been presented repeatedly (Treisman, 1960), or recall of the shadowed words turns out to be very poor (Cherry, 1953). Due to this strong evidence for the difficult processing of words presented to the to-be-ignored (i.e., unattended) channel in dichotic listening tasks, it is especially impressive that cues specified in implementation intentions are able to draw attention to the unattended channel. Apparently, cues specified in implementation intentions break through selective attention processes. As we tested the sensitivity to the specified cues in a completely different task than the one in which participants formed their implementation intentions, it can be assumed that participants were unlikely to be monitoring the appearance of target cues in the dichotic listening task. Also the observed effect seems to be rather stable over time. Note that implementation intentions were formed one day prior to performing the dichotic listening task.

One might wonder whether we did not observe the right ear advantage (REA) which is often observed in dichotic listening tasks. The REA usually is interpreted as an indicator of left hemisphere processing superiority of verbal stimuli (e.g., Gadea, Espert, & Chirivella, 1997; Sætrevik & Hugdahl, 2007). However, in studies in which the presentation of the words was preceded by primes (Sætrevik & Hugdahl, 2007), no REA was observed. Therefore, it can be assumed that if stimuli are presented that are relevant to participants (i.e., implementation intention participants in the present study) or with which participants are familiar (i.e., control participants in the present study), the REA does not occur.

## Study 2: Long-Term Memory for Specified Cues

An early and a delayed free recall test was used to examine whether the activation of the representation of cues specified in the if-component of an implementation intention is raised to a higher level. In addition, it was explored whether this heightened activation is moderated by lasting commitment. As implementation intentions are seen as being formed in the service of

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superordinated goals (Gollwitzer, 1999), we wanted to know whether this heightened activation is reduced when people disengage from the respective superordinate goal. In addition, we explored whether a person`s comittment to the formed implementation intentions can increase the activation of the specified cues.

Participants first had to form implementation intentions to be acted upon in an upcoming task (see below); different options for relevant implementation intentions were offered to select from. Thereafter, participants' commitment to their implementation intentions was manipulated by suggesting that sticking to such plans would be profitable or not. Two thirds of the participants were encouraged to commit strongly to their implementation intentions (high commitment participants), and one third was discouraged to do so (low commitment participants). Moreover, half of the high commitment participants were made to disengage from the goal to perform in the upcoming task (disengaged participants); all other participants did not receive a disengagement manipulation. Finally, all participants were asked to recall as many of the offered plans as possible, either 15 min or two days later.

We predicted that high commitment participants should recall their selected options better than non-selected, alternative options. This was not expected for low commitment and disengaged participants. Moreover, it was predicted that high commitment participants should recall specified cues better than low commitment and disengaged participants. The recall of alternative cues, however, should not vary between conditions. All of these effects were expected for both 15 min and 2 days of delay of the recall task.

**Method**

**Participants**

Male university students enrolled in different fields of study took part (n = 120; age:  $M = 25.6$ ,  $SD = 3.5$ ). We randomly invited half of them to take part in (allegedly) two experiments run on the same day, and the other half to take part in (allegedly) two experiments being run two days apart.



## Design

The study followed a 2 within (Target Words: critical vs. noncritical)  $\times$  2 between (Delay of Recall: 15 min vs. 2 days)  $\times$  3 between (Commitment: low vs. high vs. disengaged) mixed model factorial design. As the dependent variable the free recall of target words was measured 15 min or two days after implementation intentions formation.

## Procedure

Participants came into the laboratory individually. They completed a questionnaire on how they organized their everyday life. After completing this questionnaire, participants were informed that the aim of the study was to evaluate a new type of therapy geared towards increasing self-efficacy. For one of the sessions of this therapy, two games (a computer and a handicraft game) were said to be currently developed so that they can be used to increase self-efficacy. Participants were then asked whether they would like to play these games; all participants agreed and formed the goal intention “I want to play these games!” Participants were told that between introducing the games and answering the self-efficacy questionnaire they might be asked to perform another task in order to fill in the time between these two parts of the study (actually, a distraction task was handed out).

Participants planned in detail when and where to play the games. For this purpose, pairs of situational cues were presented on a computer screen one pair after the other. Altogether 9 pairs were presented; a new pair was offered whenever a selection had been made. By pressing keys, participants chose one cue of each pair in each trial thus specifying when and where they intended to play the games. We emphasized that in the present session, participants would only plan where and when they wanted to play the games, but that the games would be played later. When the planning task was completed, participants were told that they would be called again to arrange an appointment for a second session in which the games would actually be played.

Thereafter, (allegedly) based on their answers to the life organization questionnaire filled out at the beginning of the experiment, all participants were given false feedback. Two thirds were

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2 told that they qualify as somebody who benefitted from committing strongly to one`s plans (high  
3 commitment condition), whereas one third was told that they benefitted from staying flexible  
4 regarding their implementation intentions (low commitment condition). After this false feedback,  
5 participants indicated their plan commitment and the plausibility of the received feedback.  
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11 Then, a concentration test (Düker, 1953) was handed out as a distraction task.  
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13 Participants solved a series of easy but tedious arithmetic problems. To establish the  
14 disengaged condition, half of the high commitment participants were informed in the middle of  
15 this task that the play therapy games would not have to be performed anymore as enough data  
16 had been collected already. This was done to disengage these participants from the goal  
17 intention to actually play the games. After this interruption, they performed the rest of the  
18 concentration test.  
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29 At the end of the concentration test (i.e., about 15 min after forming implementation  
30 intentions), for one half of all participants an incidental recall was requested in the context of a  
31 second experiment. These participants were asked to write down as many selected *and* non-  
32 selected options of playing the two games as possible. The other half of the participants were  
33 asked to take part in an (ostensibly) second experiment conducted two days later. These  
34 participants were not informed that they would then be asked to recall which options to play  
35 the games were available to guarantee that the recall test would be incidental and that they  
36 would not try to rehearse these options in the meantime. Participants were debriefed, thanked,  
37 and paid once they had completed the recall test (five Euros for each session).  
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50 **Materials**

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52 **Life organization questionnaire.** This questionnaire (allegedly) investigated how  
53 everyday life is organized. Participants indicated how often they performed leisure activities, the  
54 time they reserved for leisure activities in a typical week, the times of each day at which they  
55 engaged in activities related to their professional and family responsibilities, and for several  
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specific activities whether they perform them at the same time each day. Finally, participants rated whether they arranged their activities on a weekday differently as compared to the weekend.

**Targets.** The situational cues offered to participants as potential if-components of their implementation intentions were presented as answers to questions for each of the two games, whereby each question was matched with two choice options (see Appendix).

**Manipulation check.** The plan commitment questionnaire asked the following questions: “How strongly are you determined to play the games under the conditions you have selected?”, „How important is it for you to play the games under the conditions you have selected?“, and “How committed are you to play the games under the conditions you have selected?” Participants marked their answers on 9 point-scales ranging from zero (“not determined”/“not important”/“not committed at all”) to 9 (“very determined”/“very important”/“strongly committed”). The perceived plausibility of the feedback was checked with the following item: “Are the results of the questionnaire on life organization in accordance with your own experiences?”

## Results

### Manipulation Check

Answers to the feedback plausibility question did not reveal significant differences between the three commitment conditions ( $M = 6.48$ ,  $SD = 2.01$ ),  $F(1,118) = 1.59$ ,  $p = .21$ ,  $\eta^2 = .01$ . This suggested that both the low and high commitment induction was equally well accepted. As the answers to the commitment items showed high intercorrelations (Cronbach’s  $\alpha = .76$ ), a commitment index was computed by taking the mean. A Commitment  $\times$  Delay of Recall ANOVA on the commitment index revealed a significant main effect of Commitment,  $F(2,114) = 7.93$ ,  $p < .001$ ,  $\eta^2 = .12$ . Comparing the means of low commitment participants ( $M = 3.29$ ,  $SD = 1.76$ ) with that of high commitment participants ( $M = 4.15$ ,  $SD = 2.06$ ) revealed that the expected difference was significant,  $t(78) = 2.00$ ,  $p < .05$ . Also, low commitment participants ( $M = 3.29$ ,  $SD = 1.76$ ) reported lower commitment than disengaged participants ( $M = 5.06$ ,  $SD = 2.06$ ),  $t(78) = 4.12$ ,  $p < .001$ .

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3 .001. This is in line with the fact that the commitment measure was taken prior to the  
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5 disengagement manipulation.  
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7 **Free Recall**  
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10 Sum scores of the correctly remembered targets in the recall test were computed. A mixed-  
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12 model 2 within (Target Words: critical vs. noncritical)  $\times$  2 between (Delay of Recall: 15 min vs.  
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14 2 days)  $\times$  3 between (Commitment: low vs. high vs. disengaged) ANOVA on these sum scores  
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16 was computed. A significant Target Words  $\times$  Commitment interaction,  $F(2,114) = 6.98, p < .01,$   
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18  $\eta^2 = .11$ , a significant main effect of Target Words,  $F(1,114) = 21.16, p < .001, \eta^2 = .16$ , and a  
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20 significant main effect of Delay of Recall was observed,  $F(1,114) = 65.98, p < .001, \eta^2 = .37$ . No  
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22 other main or interaction effects reached significance (Target Words  $\times$  Delay of Recall  $\times$   
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24 Commitment interaction,  $F < 1$ ; Target Words  $\times$  Time interaction,  $F(1,114) = 2.61, p = .11, \eta^2 =$   
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26  $.02$ ; main effect of Commitment,  $F(1,114) = 2.09, p = .13, \eta^2 = .03$ ; see Fig. 2).  
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31 Due to the significant main effect of Delay of Recall, 2 (Target Words)  $\times$  3 (Commitment)  
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33 ANOVAs on the sum scores were computed for each delay period separately. Beginning with the  
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35 recall after 15 min, a significant Target Words  $\times$  Commitment interaction,  $F(1,57) = 3.19,$   
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37  $p < .05, \eta^2 = .10$ , and a significant main effect of Target Words was observed,  $F(1,57) = 5.61, p$   
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39  $< .05, \eta^2 = .09$ . The main effect of Commitment did not reach significance,  $F(2,57) = 1.49, p =$   
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41  $.23, \eta^2 = .05$ . Planned  $t$ -tests revealed that high commitment participants showed a better recall  
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43 ( $M = 7.20, SD = 2.12$ ) of the critical targets compared to the noncritical targets ( $M = 6.20, SD =$   
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45  $1.85$ ),  $t(19) = 3.68, p < .01$ . Low commitment participants showed no better recall of the critical  
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47 targets ( $M = 5.90, SD = 1.55$ ) compared to the noncritical targets ( $M = 5.75, SD = 1.71$ ),  $t < 1$ .  
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50 The same was true for the disengaged participants (critical targets:  $M = 5.90, SD = 2.20$ ;  
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52 noncritical targets:  $M = 5.85, SD = 1.98, t < 1$ ). Thus, as early as after a delay of only 15 min low  
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54 commitment and disengaged participants were no longer able to recall the critical targets better  
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56 than noncritical targets.  
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Next, it was tested whether the critical targets were better remembered depending on commitment. It was observed that high commitment participants ( $M = 7.20$ ,  $SD = 2.12$ ) showed a better recall of the critical targets than low commitment participants ( $M = 5.90$ ,  $SD = 1.55$ ),  $t(38) = 2.21$ ,  $p < .05$ ; this was also true for disengaged participants ( $M = 5.90$ ,  $SD = 2.20$ ),  $t(38) = 1.91$ ,  $p < .05$  (*one-tailed*). There was no difference between the low commitment and the disengaged condition,  $t < 1$ . Finally, there was no difference between the commitment conditions on the recall of the noncritical target words, all  $ts < 1$ .

Analogous analyses were computed for participants' recall performance 2 days after forming their implementation intentions. A Target Words  $\times$  Commitment ANOVA on the sum scores of options remembered after two days was computed. This mixed-model ANOVA uncovered a significant Target Words  $\times$  Commitment interaction,  $F(2,57) = 3.85$ ,  $p < .05$ ,  $\eta^2 = .12$ , and a significant main effect of Target Words,  $F(1,57) = 16.02$ ,  $p < .001$ ,  $\eta^2 = .22$ . The factor Commitment did not reach significance,  $F < 1$ . The analysis of the Target Words  $\times$  Commitment interaction by means of planned  $t$ -tests revealed that high commitment participants showed a better recall of the critical targets ( $M = 4.70$ ,  $SD = 1.78$ ) compared to the noncritical targets ( $M = 3.05$ ,  $SD = 1.54$ ),  $t(19) = 5.18$ ,  $p < .001$ . However, low commitment participants showed no better recall of the critical targets compared to the noncritical targets (critical targets:  $M = 3.65$ ,  $SD = 1.87$ ; noncritical targets:  $M = 3.25$ ,  $SD = 2.15$ ),  $t(19) = 1.09$ ,  $p = .28$ . The same was true for the disengaged participants (critical targets:  $M = 3.45$ ,  $SD = 2.09$ ; noncritical targets:  $M = 3.00$ ,  $SD = 1.96$ ),  $t(19) = 1.14$ ,  $p = .27$ .

Next it was tested whether the critical targets were better remembered depending on commitment. We observed that high commitment participants showed a better recall of the critical targets ( $M = 4.70$ ,  $SD = 1.78$ ) than low commitment participants ( $M = 3.65$ ,  $SD = 1.87$ ),  $t(38) = 1.82$ ,  $p < .05$  (*one-tailed*). Furthermore, high commitment participants showed a better recall of the critical targets ( $M = 4.70$ ,  $SD = 1.78$ ) than disengaged participants ( $M = 3.45$ ,  $SD = 2.09$ ),  $t(38) = 2.04$ ,  $p < .05$ . There was no difference between the low commitment and disengaged

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3 condition,  $t < 1$ . Finally, there were no differences between the three commitment conditions on  
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5 the recall of the noncritical targets (all  $ts < 1$ ; see Figure 2).  
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8 **Discussion**

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10 The heightened accessibility of cues specified in the if-component of implementation  
11 intentions was measured by a recall test. Only participants who were highly committed to  
12 implementation intentions were better in the recall of selected cues compared to non-selected cues  
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14 after a delay of 15 min, and even after a delay of two days. However, if this commitment was low  
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16 or participants were suggested to disengage from the superordinate goal, the specified cues were  
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18 no longer remembered better than their nonspecified alternatives, and this as early as 15 min after  
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20 forming the implementation intentions.  
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26 High commitment participants were also significantly better in the recall of the specified  
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28 cues compared to low commitment participants, after a delay of 15 min and two days later.  
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30 Comparing the recall of specified cues in high commitment but disengaged from the goal  
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32 participants with high commitment but not disengaged participants, the latter significantly  
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34 outperformed the former in the recall of specified cues after a delay of 15 min and after two days.  
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36 Thus, concerning the comparison between the recall of specified and nonspecified cues,  
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38 disengaged participants showed the same recall performance as low commitment participants. It is  
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40 unlikely that this pattern of results is due to differential rehearsal as the distractor task applied  
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42 after the formation of the implementation intentions should have prevented the rehearsal.  
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44 Research on rehearsal has shown in general that the spontaneous rehearsal of information that is  
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46 not expected to be asked for in a later recall is very unlikely, especially if a distraction task is  
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48 applied immediately after the encoding of the information (e.g., Wixted, 1992).  
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55 One strength of the present study is the fact that participants did create their own  
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57 individual implementation intentions, that is the implementation intentions were not assigned by  
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59 the experimenter but were individually chosen. Murgraff, White, and Phillips (1996) as well as  
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Achtziger et al. (2008, Study 2) have also studied individually chosen implementation intentions,

and showed that they help to control unwanted responses (e.g., negative feelings) as was observed in other studies for assigned implementation intentions (e.g., Achtziger et al., 2008, Study 1). A further positive aspect of the present study is the fact that commitment to participants' plans was manipulated, and that commitment turned out to be a strong moderator of the postulated heightened activation of specified cues. Also, the present study used a disengagement manipulation to weaken the superordinate goal; again, this manipulation was found to undermine heightened activation. Thus it seems that implementation intention effects need not only have to be backed up by strong goal intentions (Sheeran, Webb, & Gollwitzer, 2005, Study 1) and a situational context that activates the superordinate goal (Cohen, Bayer, Jaudas, & Gollwitzer, Study 1, 2008; Sheeran et al., 2005, Study 2). Rather, for implementation intentions to unfold their effects it needs to be assured that people have not yet given up their striving for the superordinate goal (i.e., have not yet disengaged from this goal).

This goal disengagement finding of the present Study 2 is in line with research on the ISE (Goschke & Kuhl, 1993). For instance, Marsh et al. (1998) observed in a lexical decision task that was conducted after an intention had been fulfilled or cancelled, that memory for this intention became inhibited relative to material that was not associated with the intention.

## General Discussion

Forming implementation intentions by specifying cues as a condition for initiating goal-directed responses leads to goal attainment more often than setting mere goal intentions (reviews by Achtziger & Gollwitzer, 2008; Gollwitzer & Sheeran, 2006), even when goal-directed behaviors are inconvenient (e.g., Gollwitzer & Brandstätter, 1997) or unpleasant (e.g., Orbell & Sheeran, 2000). Implementation intentions are also successful in shielding one's goal-directed behavior (Achtziger et al., 2008), and in strengthening self-efficacy (Bayer & Gollwitzer, 2007).

As a consequence of forming an implementation intention, a mental link is created between a specified cue and the goal-directed behavior (Gollwitzer, 1993, 1999). This link is expected to facilitate goal attainment as follows: By forming an implementation intention, the

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representation of the specified cue becomes highly activated and therefore more accessible (Aarts, Dijksterhuis, & Midden, 1999). The initiation of the response specified in the then-component is also facilitated: The specified response is executed immediately (Gollwitzer & Brandstätter, 1997), efficiently (Brandstätter et al., 2001), and without requiring further conscious intent (Bayer et al., 2009).

We tested whether the postulated heightened activation of selected cues results in specific attention and long-term memory processes. In Study 1, attention processes associated with the specified cues were investigated by means of a dichotic listening task. In dichotic listening tasks it is assumed that since our attention is limited, it is difficult to spread attention to all information processing channels at once; one is only able to effectively attend to one channel at a time (e.g., Broadbent, 1954; Bryden, 1988). In Study 1, we presented words to the unattended channel that described individually specified cues. Simultaneously, participants repeated semantically unrelated words presented to the attended channel. Compared to control participants, implementation intention participants responded slower to the words presented to the attended channel if specified cues were presented to the unattended channel. This suggests that despite being strongly distracted, specified cues still attract attention; implicating that even when being strongly involved in other tasks, cues selected for enacting goal-directed behavior will catch attention.

Study 2 tested whether the heightened activation of specified cues is also indicated by a better long-term memory of these cues compared to alternative cues. It was observed that both after 15 min and after two days, specified cues were better remembered than alternative cues. These effects were moderated by the commitment to stick to one's action plans and by holding the superordinated goal. Participants with low commitment did neither show a memory advantage for the specified cues after 15 min nor after two days. Also, participants who no longer held the superordinate goal did not show a better memory for specified cues compared to non-chosen



alternatives and this even though commitment to the implementation intentions had been induced to be high.

In line with this reasoning, Webb and Sheeran (2007) observed tht it is the strength of the link between the if (cue) and the then (goal-directed response) component of implementation intentions that mediates their effects.

One strength of Study 2 lies in the manipulation of the length of the retention interval. Only a few studies investigated whether memory for intentions is influenced by the delay between forming the intention and its execution. For instance, Hicks, Marsh and Russel (2000) investigated in 5 event-based prospective memory studies the importance of the retention interval. They varied the retention interval (2.5 min vs. 15 min) during which participants performed a distracter activity before their prospective memory was examined. Memory for intentions revealed to be better after a delay compared to immediate retrieval which was in contrast to retrospective memory studies in which usually a decline of performance with an increased retention interval is observed (Ebbinghaus, 1885/1964). Other experiments (e.g., Einstein, Holland, McDaniel, & Guynn, 1992; Guynn, McDaniel, & Einstein, 1998) observed no significant differences in memory as a function of retention interval (e.g., 15 min vs. 30 min). In summary, these findings on the effects of the length of the retention intervals are quite mixed.

Going beyond these former studies, Study 2 of the present paper varied the time interval between the encoding of the cues and the recall test by using a 15 min and a 2 days condition. As we did not announce that participants in the two days condition would be asked which cues they had specified one can assume that they were not motivated to use external aids or memorization strategies to achieve the observed heightened recall performance for selected cues. Accordingly, we can assume that the observed strong memory advantage for cues specified in the if-components of implementation intentions is primarily due to a state of heightened activation that persisted for two days after their formation.

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*Alternative Explanation of the Intention Superiority Effect*

Besides the state of a heightened activation of intentions, the ISE can be explained in terms of spontaneous activation of motor and sensory information for verbal materials that are intended for future execution (Freeman & Ellis, 2003; Koriat et al., 1990; McDaniel & Einstein, 2007). A verbal script like “washing” might activate motor and sensory information associated with the action of washing; whereby the sensory-motor information and associated processing might strengthen the accessibility and retention performance of the intention. This argument is supported by Koriat et al. (1990) who showed that participants who encoded simple noun-verb phrases for future enactment remembered these phrases better than noun-verb phrases that were encoded for verbal report. Furthermore, Freeman and Ellis (2003) argued that if the ISE were only based on a privileged status of the intention, the modality of interference should not affect it. However, they observed that the ISE can be eliminated by motor interference tasks but not by verbal interference tasks. It seems possible that the encoding of action cues is also enriched with sensorimotor information as participants want to enact their plans in the future (see Badet et al., 2006). Thus, the heightened activation of cues specified in implementation intentions may be caused by the anticipated enactment.

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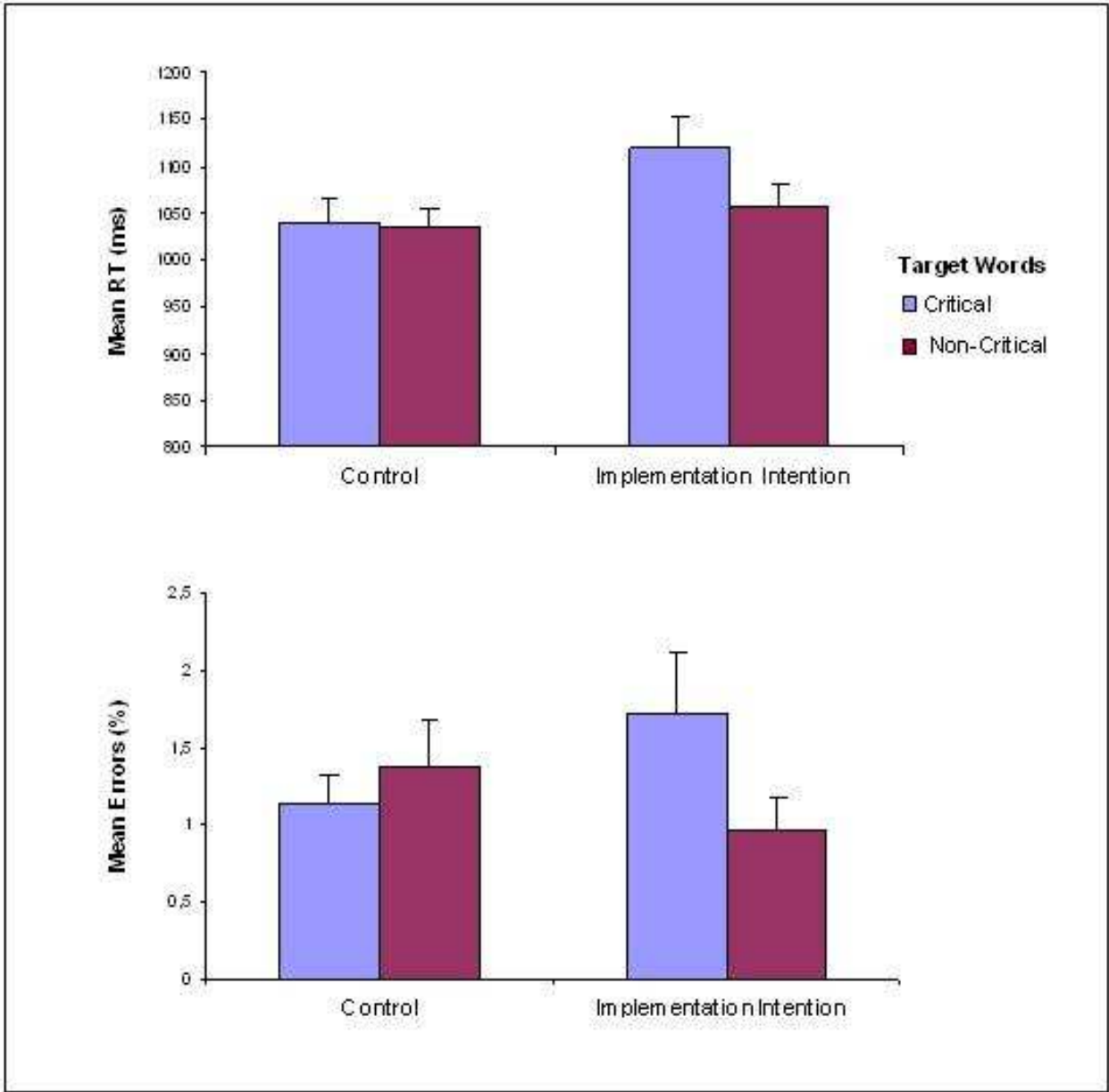
Figure Caption

*Figure 1.* Mean Latencies and Errors on the Critical and Non-Critical Targets by Intention  
(Study 1).

*Figure 2.* Mean Numbers of Correctly Remembered Targets by Plan Commitment and Delay  
(Study 2).

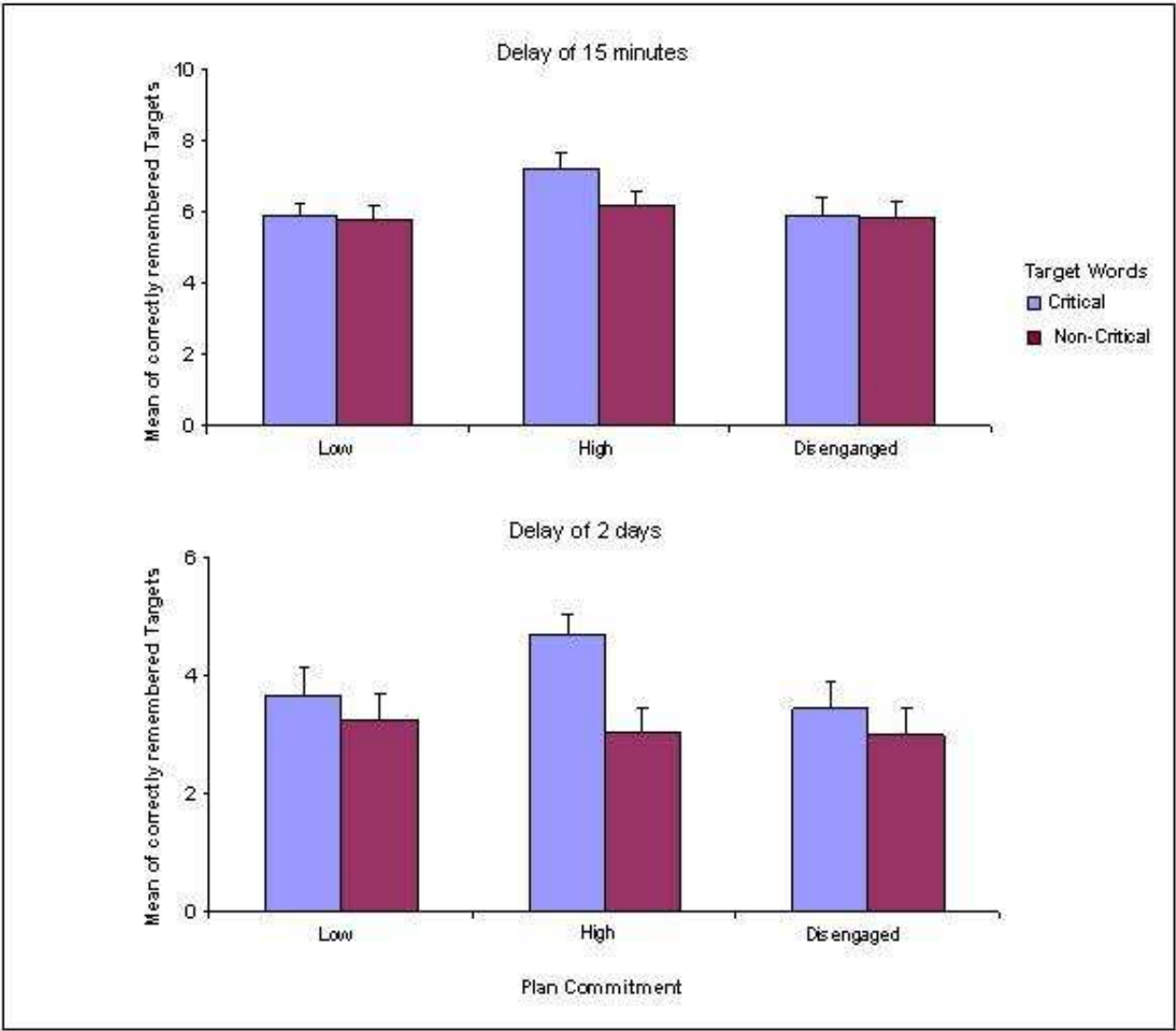


Figure 1. Mean Latencies and Errors for Critical and Noncritical Targets by Intention (Study 1).



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Figure 2. Mean Number of Correctly Remembered Targets by Commitment and Delay (Study 2).



### *Appendix List of Options*

“Where will you play the games?” (Game 1: a. laboratory, b. home office/Game 2: a. studio, b. gym), “Together with whom?” (Game 1: a. alone, b. experimenter/Game 2: a. another student, b. staff of the institute), “When will you play it?” (Game 1: a. Monday, b. Thursday/Game 2: a. Tuesday, b. Friday), “Which kind of equipment will you use?” (Game 1: a. personal computer, b. laptop/Game 2: a. workbench, b. table for making handicrafts), “Which kind of tools will you use?” (Game 1: a. mouse, b. joystick/Game 2: a. threads, b. balls), “Which kind of recording tool will you ask for?” (Game 1: a. camcorder, b. one-way mirror/Game 2: a. interview, b. self-report).